

Impact of Improving Lead Screening Mandates for Children Aged 1-3 in Georgia

Health Impact of Lead Exposure for Children:

Lead poisoning remains a serious and invisible threat to children's health and development. The CDC's reference level of lead in the bloodstream for children is 3.5 µg/dL¹. Even at low levels, lead exposure can impair cognitive function, lower IQ, disrupt language and behavior, and contribute to learning disabilities and attention deficit behavior in children. Higher levels of lead exposure pose even greater risks and can lead to seizures, nervous system damage, and, in extreme cases, coma or death¹.

Lead Issue in Georgia:

In 2019, a survey in Georgia found that 41% of children ages 2-7 had blood lead levels at or above 5 µg/dL¹. Exposure in Georgia disproportionately affects children in older housing, low-income communities, and neighborhoods where housing conditions and historic lead paint use remain common.

Importance of Early Childhood Blood Lead Screening:

This analysis examines the potential impact of expanding early childhood blood lead screening in Georgia. Because lead exposure is typically asymptomatic, blood lead screening is the only reliable way to identify children at risk before irreversible harm occurs. Early detection can reduce future healthcare and developmental costs.

Rationale for Targeting Ages 1–3 for Screenings:

Children between the ages of 1-3 are the focus of this analysis because they are biologically the most vulnerable to lead exposure and consistently show the highest blood lead levels². They are especially vulnerable due to hand-to-mouth behavior and rapid nervous system development³. National guidelines recommend routine screening at 12 and 24 months for this reason⁴. Most states with lead screening mandates for children and target this same age range, making it the most evidence-based and widely adopted window for early detection.

Types of Lead Screening Mandates for Children:

States vary widely in how they structure lead screening requirements. For children not enrolled in Medicaid, states usually use a universal, targeted or hybrid screening system. Universal screening requires all young children regardless of risk factors to receive blood lead tests at specified ages⁵. Targeted screening limits mandatory testing to children who meet specific risk criteria, such as living in housing built before 1978 or having known environmental exposure⁶. Targeted screening requires fewer tests, but it often misses high-risk children because families may not know their housing age and providers applying risk questionnaires inconsistently⁶. Hybrid screening policies combine both approaches by requiring universal testing in clearly defined high risk areas, such as certain counties, while applying targeted screening in lower risk regions.

Why Universal Screening is Best for Georgia:

For Georgia's context, a universal structure would provide the most benefit. This approach maximizes case findings, particularly in states where risk is more widespread and has a high ratio of old housing, and it reduces the likelihood that high-risk children are missed due to inconsistent or inaccurate risk assessments. The goal of the universal mandate is to increase early detection of elevated blood lead levels among children in Georgia ages 1–3 by implementing a universal screening mandate.

Current Screening Policies in Georgia:

Georgia uses a targeted screening system in which only Medicaid-enrolled children are guaranteed mandatory screenings at 12 and 24 months. All other children are tested only if they meet specific risk criteria outlined by the local department of health⁷. Risk criteria can include factors such as living in a home, attending a school or daycare facility built prior to 1978 or exposure to an adult whose job or hobbies involve lead⁸. Under this system, lead screening is recommended based on health department guidance, but it is not required for all children at 12 and 24 months unless they are covered by Medicaid.

New Jersey as a Model Comparison:

New Jersey offers a strong model for Georgia because it mandates blood lead screenings at 12 and 24 months and risk assessments at wellness visits up to 6 years of age⁹. New Jersey has consistently been ranked among the top states for childhood lead screening coverage¹⁰. In 2022, 78% of New Jersey children received at least one blood lead test by 3 years old, compared with only 30% in Georgia, demonstrating the impact of a universal mandate^{12,13}. Laws such as these have led to increased identification of lead exposure in asymptomatic children, allowing for quicker action and medical intervention. Using New Jersey's documented screening performance provides Georgia with a realistic benchmark for estimating the public health and economic gains of adopting a universal screening mandate.

Methods and Results:

In my analysis, I estimated the impact of increasing childhood lead screening coverage under a universal testing mandate on elevated blood lead level case detection. I estimated costs associated with Georgia adopting a universal screening mandate for children ages 1–3 by estimated the cost per test and identifying approximately how many additional tests would be administered under this mandate. The number of eligible children (370,000) was based on Georgia population data¹⁰. Georgia's current screening rate (30%) and EBLL prevalence (3.8%) were taken from state surveillance estimates^{12,14}. New Jersey achieves 80% screening coverage for children 3 and under and was used as a realistic benchmark for projecting Georgia's screening rate under a universal mandate¹³. The prevalence of EBLL was assumed constantly across scenarios. At an estimated cost of \$30 per blood lead test, this model projects approximately 185,000 additional screenings in Georgia, totaling \$5,550,000¹⁵. Using the EBLL rate, roughly 7,030 additional cases would be identified. Identifying these cases through increased screenings would allow for earlier intervention, which can reduce the potential progression of lead exposure and associated health and developmental harm.

Table 1-Elevated Blood Lead Level Detection (EBLL) & Cost of Additional Screenings

Variable (V)	GA Current Estimate	Projected Estimate	Calculation / Explanation
1. Eligible Children ¹⁰	370,000	370,000	Proportional age distribution
2. Screening Rate ^{12,13}	30%	80%	Projected estimate based on NJ
3. Children Tested Annually	111,000	296,000	(V1) *(V2)
4. * EBLL Rate ($\geq 3.5\mu\text{g/dL}$) ¹⁴	3.80%	3.80%	(EBLL)/ (total screenings) *100
5. Number of EBLL Cases	4,218	11,248	(V3) *(V4)
6. Additional Screenings	185,000		296,000 - (V3)
7. Additional EBLL	7,030		11,248 - (V5)
8. Cost per Screening ¹⁵	\$30	\$5,550,000	(V6) *(V8)

**EBLL=Elevated Blood lead level*

Sensitivity Analysis:

My sensitivity analysis evaluates potential economic benefits that could result from increased detection of elevated blood lead levels under a universal screening mandate in Georgia, using evidence from New Jersey as a benchmark. A New Jersey Department of the Public Advocate report notes that “net societal benefits arising from these improvements in high school graduation rates... would amount to \$31,000 per child”, highlighting that societal outcomes such as education costs are included in economic considerations¹⁶. In this analysis, the estimated economic benefits reflecting the impact of additional EBLL cases detected through expanded early screening and the early interventions they lead to, which can reduce the potential progression of lead exposure and its developmental harms. These estimates do not assume complete elimination of environmental lead exposure or full abatement of future cases, focusing instead on realistic, evidence informed benefits of early detection.

Table 2 - One way Sensitivity Analysis: Adjusted Cost Savings per Additional EBLL Case Identified

Scenario	Cost Savings per EBLL Case	Additional EBLL Cases	Economic Benefit
High	\$30,000 ¹⁶	7,030	\$210,900,000
Medium	\$15,000	7,030	\$105,450,000
Low	\$5,000	7,030	\$35,150,000

Conclusion:

This analysis assessed the impact of a universal blood lead screening mandate for children ages 1–3 in Georgia. Based on current screening rates and elevated blood lead levels, and using New Jersey as a guide, I estimated that moving to a universal mandate could yield 185,000 additional screenings costing about \$5,550,000 and identify 7,030 more children with EBLL annually. Early detection enables timely interventions, mitigating developmental and health consequences. The sensitivity analysis, based on New Jersey’s documented outcomes, and indicates that the economic benefits of these additional detections could range from \$35 million to \$210 million, reflecting health and educational impacts without assuming full abatement of lead exposure. These findings suggest that universal screening offers tangible public health and economic advantages and offers ground to support informed policy decisions to protect vulnerable children aged 1-3 in Georgia under a universal mandate.

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